Study Notes: Chemistry - G9

UNIT 1: CHEMISTRY AND ITS IMPORTANCE: 1.1 Definition and Scope of Chemistry, 1.2 Relationship between Chemistry and Other Natural Sciences, 1.3 The Role Chemistry Plays in Production and in the Society, 1.4 Some Common Chemical Industries in Ethiopia, UNIT 1: Unit Summary

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Chemistry and its Importance

This chapter introduces the fundamental science of chemistry, its core definitions, various branches, and its wide-ranging impact on our daily lives, society, and industry, including specific examples from Ethiopia.

1.1 Definition and Scope of Chemistry

This section delves into the fundamental definition of chemistry and explores its extensive areas of study and application.

1.1.1 Definition of Chemistry

Chemistry is a foundational science that explores the composition, properties, and changes of matter.

Chemistry: The science that deals with the properties, composition, and structure of substances (elements and compounds), the transformations they undergo, and the energy that is released or absorbed during these processes.

To understand chemistry, it's crucial to define its core concepts:

- **Substance:** A particular kind of matter with uniform properties.
  - Examples: Gold, silver, water, soap, table salt.
- **Matter:** A physical substance; that which occupies space and possesses rest mass.
  - Examples: Book, pencil, television, stool.
- **Property of a Substance:** Its attribute, quality, or characteristic.
  - Every substance has unique properties, allowing us to distinguish it from others.
  - These unique properties stem from a substance's unique composition and structure.
  - Example: Water has no color, taste, or shape.
- **Composition:** The nature of something's ingredients or constituents; how a whole or mixture is made up.
  - Example 1: Table salt is chemically composed of the elements sodium and chlorine (NaCl).
  - Example 2: Stainless-steel spoons are a solid solution (alloy) of chromium, carbon, and other elements.
- **Structure:** The arrangement and relationships between the parts or elements of something complex.
  - Example: The structure of a school building is defined by the arrangement of its roof, ceiling, doors, windows, walls, and floor.
- **Transformation of a Substance:** A marked change in form, nature, or appearance.
  - Substances continuously change due to both external and internal forces.
  - These transformations are always accompanied by energy changes (released or absorbed).

Key Takeaway: Chemistry is the study of what matter is made of, how it behaves, and how it changes, including the energy involved in those changes.

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📝 Practice Question 1 / የልምምድ ጥያቄ 1 (EUEE-style)

Question / ጥያቄ: Which of the following best defines chemistry? / ከሚከተሉት ውስጥ ኬሚስትሪን በተሻለ ሁኔታ የሚተረጉመው የቱ ነው?

Options / አማራጮች:

- A. The study of living things [የሕይወት ያላቸው ነገሮች ጥናት]
- B. The science that deals with the properties, composition and structure of substances, the transformations they undergo, and the energy released or absorbed [የንጥረ ነገሮችን ባህሪያት፣ ስብጥር እና አወቃቀር፣ የሚያልፉባቸውን ለውጦች እንዲሁም የሚለቀቀውን ወይም የሚወሰደውን ኃይል የሚያጠና ሳይንስ]
- C. The study of rocks and the earth [የድንጋዮችና የምድር ጥናት]
- D. The science concerned only with the nature and properties of matter and energy [የቁስንና የኃይልን ተፈጥሮና ባህሪ ብቻ የሚመለከት ሳይንስ]

Correct Answer / ትክክለኛ መልስ: B

Explanation (English): Chemistry covers the properties, composition and structure of substances, their transformations, and the energy changes involved. Option D is the main distractor: it describes physics and leaves out composition, structure and transformations.

ማብራሪያ (አማርኛ): ኬሚስትሪ የንጥረ ነገሮችን ባህሪያት፣ ስብጥር እና አወቃቀር፣ የሚያልፉባቸውን ለውጦች እና የሚኖረውን የኃይል ለውጥ ያጠናል። አማራጭ D ዋናው አሳሳች መልስ ነው፤ ይህ የፊዚክስ ትርጉም ሲሆን ስብጥርን፣ አወቃቀርንና ለውጥን አያካትትም።

📝 Practice Question 2 / የልምምድ ጥያቄ 2 (EUEE-style)

Question / ጥያቄ: Which of the following is an example of a *substance* (a particular kind of matter with uniform properties)? / ከሚከተሉት ውስጥ ንጥረ ነገር (ወጥ ባህሪ ያለው የአንድ ዓይነት ቁስ) ምሳሌ የሆነው የቱ ነው?

Options / አማራጮች:

- A. Book [መጽሐፍ]
- B. Pencil [እርሳስ]
- C. Gold [ወርቅ]
- D. Television [ቴሌቪዥን]

Correct Answer / ትክክለኛ መልስ: C

Explanation (English): Gold, silver, water, soap and table salt are examples of substances. Book, pencil and television are the main distractors: they are listed as examples of matter, that is, objects that occupy space and have mass.

ማብራሪያ (አማርኛ): ወርቅ፣ ብር፣ ውሃ፣ ሳሙና እና የገበታ ጨው የንጥረ ነገር ምሳሌዎች ናቸው። መጽሐፍ፣ እርሳስና ቴሌቪዥን ቦታ የሚይዙና ክብደት ያላቸው የቁስ ምሳሌዎች ናቸው፤ ስለዚህ እነዚህ አሳሳች አማራጮች ናቸው።

📝 Practice Question 3 / የልምምድ ጥያቄ 3 (EUEE-style)

Question / ጥያቄ: “Table salt is chemically composed of the elements sodium and chlorine (NaCl).” This statement describes the _____ of table salt. / “የገበታ ጨው በኬሚካላዊ መልኩ ከሶዲየምና ከክሎሪን ንጥረ ነገሮች (NaCl) የተሠራ ነው።” ይህ ዐረፍተ ነገር የገበታ ጨውን _____ ይገልጻል።

Options / አማራጮች:

- A. Composition [ስብጥር]
- B. Structure [አወቃቀር]
- C. Transformation [ለውጥ]

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- D. Matter [ቁስ]

Correct Answer / ትክክለኛ መልስ: A

Explanation (English): Composition tells what ingredients or constituents something is made of, and NaCl shows that salt is made of sodium and chlorine. Structure is the main distractor: it describes how the parts are arranged, not what they are.

ማብራሪያ (አማርኛ): ስብጥር አንድ ነገር ከምን ከምን እንደተሠራ ይገልጻል፤ NaCl ጨው ከሶዲየምና ከክሎሪን እንደተሠራ ያሳያል። አወቃቀር የክፍሎችን አቀማመጥ እንጂ ክፍሎቹ ምን እንደሆኑ ስለማይነግር አሳሳች አማራጭ ነው።

📝 Practice Question 4 / የልምምድ ጥያቄ 4 (EUEE-style)

Question / ጥያቄ: The way the roof, ceiling, doors, windows, walls and floor of a school building are arranged is an example of _____. / የአንድ ትምህርት ቤት ሕንፃ ጣሪያ፣ ጣሪያ ሥር ያለው ወለል (ሲሊንግ)፣ በሮች፣ መስኮቶች፣ ግድግዳዎችና ወለል የተደራጁበት መንገድ የ_____ ምሳሌ ነው።

Options / አማራጮች:

- A. Composition [ስብጥር]
- B. Property [ባህሪ]
- C. Transformation [ለውጥ]
- D. Structure [አወቃቀር]

Correct Answer / ትክክለኛ መልስ: D

Explanation (English): Structure is the arrangement and relationship between the parts of something complex, and the school building example in the guide illustrates exactly this. Composition is the main distractor because it deals with the ingredients, not their arrangement.

ማብራሪያ (አማርኛ): አወቃቀር የአንድ ውስብስብ ነገር ክፍሎች አደረጃጀትና ግንኙነት ነው፤ የትምህርት ቤቱ ሕንፃ ምሳሌም ይህንኑ ያሳያል። ስብጥር የሚመለከተው ክፍሎቹን እንጂ አደረጃጀታቸውን ስላልሆነ አሳሳች አማራጭ ነው።

📝 Practice Question 5 / የልምምድ ጥያቄ 5 (EUEE-style)

Question / ጥያቄ: Which statement about the transformation of a substance is correct? / ስለ ንጥረ ነገር ለውጥ የሚከተሉት መግለጫዎች ውስጥ ትክክለኛው የቱ ነው?

Options / አማራጮች:

- A. Substances change only because of external forces [ንጥረ ነገሮች የሚለወጡት በውጫዊ ኃይሎች ምክንያት ብቻ ነው]
- B. Transformations are always accompanied by energy changes, either released or absorbed [ለውጦች ሁልጊዜ የኃይል ለውጥ (መለቀቅ ወይም መወሰድ) ይኖራቸዋል]
- C. Transformations never involve energy [ለውጦች ኃይልን በፍጹም አያካትቱም]
- D. A transformation does not change the form, nature or appearance of a substance [ለውጥ የንጥረ ነገርን ቅርጽ፣ ተፈጥሮ ወይም መልክ አይለውጥም]

Correct Answer / ትክክለኛ መልስ: B

Explanation (English): The guide states that transformations are always accompanied by energy changes, released or absorbed. Option A is the main distractor, since substances change due to both external and internal forces, not external forces alone.

ማብራሪያ (አማርኛ): መመሪያው ለውጦች ሁልጊዜ ከኃይል መለቀቅ ወይም መወሰድ ጋር እንደሚታጀቡ ይገልጻል። አማራጭ A አሳሳች ነው፤ ምክንያቱም ንጥረ ነገሮች በውጫዊም በውስጣዊም ኃይሎች ይለወጣሉ እንጂ በውጫዊ ኃይል ብቻ አይደለም።

1.1.2 Scope of Chemistry

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The study of chemistry is vast, encompassing many specialized areas and influencing nearly every aspect of life.

The study of modern chemistry is generally divided into five main disciplines:

i. Physical Chemistry: - **Definition:** The study of macroscopic properties, atomic properties, and phenomena in chemical systems. - **Focus:** Rates of chemical reactions, energy transfers in reactions, physical structure of materials at the molecular level.

ii. Organic Chemistry: - **Definition:** The study of substances containing carbon. - **Significance:** Carbon is abundant and can form over twenty million known chemicals. Most chemicals in living organisms are carbon-based.

iii. Inorganic Chemistry: - **Definition:** The study of substances that are not primarily based on carbon. - **Occurrence:** Commonly found in rocks and minerals. - **Current Importance:** Design and properties of materials for energy and information technology.

iv. Analytical Chemistry: - **Definition:** The study of the composition of matter. - **Focus:** Separating, identifying, and quantifying chemicals in samples. - **Methods:** Often uses complex instruments to determine components of unknown materials.

v. Biochemistry: - **Definition:** The study of chemical processes that occur in living things. - **Scope:** From basic cellular processes to understanding disease states for developing better treatments.

All these disciplines involve: - Taking measurements. - Making observations. - Using observations to draw conclusions. - Looking for patterns in how substances behave.

Broad Scope and Impact of Chemistry: - Because all living and non-living things are made of matter, chemistry affects all aspects of life and most natural events. - It helps in: - Explaining the natural world. - Preparing people for career opportunities. - Producing informed citizens. - Key areas where chemistry plays a role include: - Agriculture - Medicine - Food production - Building construction

Potential Negative Impacts of Chemistry: - While providing useful substances, chemistry can also result in dangerous substances. - These can negatively affect human life and the environment. - Examples: Fluorochlorohydrocarbons, oxides of nitrogen, carbon, and sulfur.

📝 Practice Question 6 / የልምምድ ጥያቄ 6 (EUEE-style)

Question / ጥያቄ: Modern chemistry is generally divided into how many main disciplines? / ዘመናዊ ኬሚስትሪ በአጠቃላይ በስንት ዋና ዘርፎች ይከፈላል?

Options / አማራጮች:

- A. Five [አምስት]
- B. Three [ሦስት]
- C. Four [አራት]
- D. Seven [ሰባት]

Correct Answer / ትክክለኛ መልስ: A

Explanation (English): The guide divides modern chemistry into five main disciplines: physical, organic, inorganic, analytical and biochemistry. Seven is the main distractor, because the guide never divides chemistry into seven branches.

ማብራሪያ (አማርኛ): መመሪያው ዘመናዊ ኬሚስትሪን በአምስት ዋና ዘርፎች ይከፍላል፤ እነሱም ፊዚካል፣ ኦርጋኒክ፣ ኢ-ኦርጋኒክ፣ ትንተናዊ ኬሚስትሪ እና ባዮኬሚስትሪ ናቸው። መመሪያው ኬሚስትሪን በሰባት ዘርፎች ስለማይከፍል ሰባት አሳሳች አማራጭ ነው።

📝 Practice Question 7 / የልምምድ ጥያቄ 7 (EUEE-style)

Question / ጥያቄ: Which branch of chemistry studies substances that contain carbon? / የትኛው የኬሚስትሪ ዘርፍ ካርቦን የያዙ ንጥረ

ነገሮችን ያጠናል?

Options / አማራጮች:

- A. Inorganic chemistry [ኢ-ኦርጋኒክ ኬሚስትሪ]
- B. Analytical chemistry [ትንተናዊ ኬሚስትሪ]
- C. Physical chemistry [ፊዚካል ኬሚስትሪ]
- D. Organic chemistry [ኦርጋኒክ ኬሚስትሪ]

Correct Answer / ትክክለኛ መልስ: D

Explanation (English): Organic chemistry is the study of substances containing carbon, and most chemicals in living organisms are carbon-based. Inorganic chemistry is the main distractor because it studies substances that are not primarily based on carbon.

ማብራሪያ (አማርኛ): ኦርጋኒክ ኬሚስትሪ ካርቦን የያዙ ንጥረ ነገሮችን ያጠናል፤ በሕይወት ፍጥረታት ውስጥ ያሉ አብዛኞቹ ኬሚካሎችም በካርቦን ላይ የተመሠረቱ ናቸው። ኢ-ኦርጋኒክ ኬሚስትሪ በዋናነት በካርቦን ላይ ያልተመሠረቱ ንጥረ ነገሮችን ስለሚያጠና አሳሳች አማራጭ ነው።

📝 Practice Question 8 / የልምምድ ጥያቄ 8 (EUEE-style)

Question / ጥያቄ: A chemist uses complex instruments to separate, identify and quantify the chemicals in a sample of unknown material. Which branch of chemistry is this? / አንድ ኬሚስት ውስብስብ መሣሪያዎችን በመጠቀም ያልታወቀ ናሙና ውስጥ ያሉ ኬሚካሎችን ይነጥላል፣ ይለያል፣ መጠናቸውንም ይወስናል። ይህ የየትኛው የኬሚስትሪ ዘርፍ ሥራ ነው?

Options / አማራጮች:

- A. Organic chemistry [ኦርጋኒክ ኬሚስትሪ]
- B. Analytical chemistry [ትንተናዊ ኬሚስትሪ]
- C. Inorganic chemistry [ኢ-ኦርጋኒክ ኬሚስትሪ]
- D. Physical chemistry [ፊዚካል ኬሚስትሪ]

Correct Answer / ትክክለኛ መልስ: B

Explanation (English): Analytical chemistry studies the composition of matter by separating, identifying and quantifying chemicals, often with complex instruments. Physical chemistry is the main distractor, because it focuses on reaction rates, energy transfers and the physical structure of materials.

ማብራሪያ (አማርኛ): ትንተናዊ ኬሚስትሪ የቁስን ስብጥር የሚያጠናው ኬሚካሎችን በመነጠል፣ በመለየትና በመጠን በመወሰን ሲሆን ብዙ ጊዜ ውስብስብ መሣሪያዎችን ይጠቀማል። ፊዚካል ኬሚስትሪ የምላሽ ፍጥነት፣ የኃይል ሽግግርና የቁሶችን አካላዊ አወቃቀር ስለሚያተኩር አሳሳች አማራጭ ነው።

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📝 Practice Question 9 / የልምምድ ጥያቄ 9 (EUEE-style)

Question / ጥያቄ: The design and properties of materials for energy and information technology is a current area of importance in which branch of chemistry? / ለኃይልና ለመረጃ ቴክኖሎጂ የሚያገለግሉ ቁሶችን ንድፍና ባህሪ ማጥናት በአሁኑ ጊዜ በየትኛው የኬሚስትሪ ዘርፍ አስፈላጊ ነው?

Options / አማራጮች:

- A. Organic chemistry [ኦርጋኒክ ኬሚስትሪ]
- B. Biochemistry [ባዮኬሚስትሪ]
- C. Inorganic chemistry [ኢ-ኦርጋኒክ ኬሚስትሪ]
- D. Analytical chemistry [ትንተናዊ ኬሚስትሪ]

Correct Answer / ትክክለኛ መልስ: C

Explanation (English): The guide lists the design and properties of materials for energy and information technology as the current importance of inorganic chemistry. Organic chemistry is the main distractor because it deals with carbon-containing substances, not this area.

ማብራሪያ (አማርኛ): *መመሪያው ለኃይልና ለመረጃ ቴክኖሎጂ የሚያገለግሉ ቁሶችን ንድፍና ባህሪ የኢ-ኦርጋኒክ ኬሚስትሪ የአሁን ጠቀሜታ አድርጎ ይዘረዝራል። ኦርጋኒክ ኬሚስትሪ ካርቦን የያዙ ንጥረ ነገሮችን ስለሚያጠና አሳሳች አማራጭ ነው።*

📝 Practice Question 10 / የልምምድ ጥያቄ 10 (EUEE-style)

Question / ጥያቄ: Which of the following are given as examples of dangerous substances that can negatively affect human life and the environment? / ከሚከተሉት ውስጥ በሰው ሕይወትና በአካባቢ ላይ አሉታዊ ተጽዕኖ ሊያሳድሩ የሚችሉ አደገኛ ንጥረ ነገሮች ምሳሌ ተደርገው የተሰጡት የትኞቹ ናቸው?

Options / አማራጮች:

- A. Fluorochlorohydrocarbons and oxides of nitrogen, carbon and sulfur [ፍሎሮክሎሮሃይድሮካርቦኖችና የናይትሮጂን፣ የካርቦንና የሰልፈር ኦክሳይዶች]
- B. Urea, ammonium sulphate and sodium nitrate [ዩሪያ፣ አሞኒየም ሰልፌት እና ሶዲየም ናይትሬት]
- C. Glass, steel and cement [*መስታወት፣ ብረት እና ሲሚንቶ*]
- D. Penicillin, cisplatin and AZT [ፔኒሲሊን፣ ሲስፕላቲን እና ኤዝዶቲ (AZT)]

Correct Answer / ትክክለኛ መልስ: A

Explanation (English): The guide gives fluorochlorohydrocarbons and oxides of nitrogen, carbon and sulfur as examples of dangerous substances from chemistry. Option B is the main distractor: urea, ammonium sulphate and sodium nitrate are chemical fertilizers that help increase crop yield.

ማብራሪያ (አማርኛ): *መመሪያው ፍሎሮክሎሮሃይድሮካርቦኖችንና የናይትሮጂን፣ የካርቦንና የሰልፈር ኦክሳይዶችን የአደገኛ ንጥረ ነገሮች ምሳሌ አድርጎ ይሰጣል። አማራጭ B አሳሳች ነው፤ ዩሪያ፣ አሞኒየም ሰልፌትና ሶዲየም ናይትሬት የሰብል ምርትን የሚጨምሩ የኬሚካል ማዳበሪያዎች ናቸው።*

📝 Practice Question 11 / የልምምድ ጥያቄ 11 (EUEE-style)

Question / ጥያቄ: Which of the following statements is NOT correct? / ከሚከተሉት መግለጫዎች ውስጥ ትክክል ያልሆነው የትኛው ነው?

Options / አማራጮች:

- A. Organic chemistry studies substances containing carbon [ኦርጋኒክ ኬሚስትሪ ካርቦን የያዙ ንጥረ ነገሮችን ያጠናል]
- B. Most chemicals in living organisms are carbon-based [በሕይወት ፍጥረታት ውስጥ ያሉ አብዛኞቹ ኬሚካሎች በካርቦን ላይ የተመሠረቱ ናቸው]
- C. Inorganic substances are commonly found in rocks and minerals [ኢ-ኦርጋኒክ ንጥረ ነገሮች በብዛት በድንጋዮችና በማዕድናት ውስጥ ይገኛሉ]
- D. Analytical chemistry mainly studies the chemical processes that occur in living things [ትንተናዊ ኬሚስትሪ በዋናነት በሕይወት ፍጥረታት ውስጥ የሚከሰቱ ኬሚካላዊ ሂደቶችን ያጠናል]

Correct Answer / ትክክለኛ መልስ: D

Explanation (English): The study of chemical processes in living things belongs to biochemistry, while analytical chemistry studies the composition of matter. Options A, B and C are all true statements taken from the guide.

ማብራሪያ (አማርኛ): *በሕይወት ፍጥረታት ውስጥ የሚከሰቱ ኬሚካላዊ ሂደቶችን ማጥናት የባዮኬሚስትሪ ሥራ ነው፤ ትንተናዊ ኬሚስትሪ ደግሞ የቁስን ስብጥር ያጠናል። አማራጮች A፣ B እና C ከመመሪያው የተወሰዱ ትክክለኛ መግለጫዎች ናቸው።*

1.2 The Relationship Between Chemistry and Other Natural Sciences

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Chemistry is often called the “central science” due to its fundamental role and extensive connections with other scientific disciplines.

Science: The process by which we learn about the natural universe by observing, testing, and then generating models that explain our observations.

The physical universe is vast, leading to many different branches of science: - **Biology:** The study of living things. - **Geology:** The study of rocks and the earth. - **Physics:** The branch of science concerned with the nature and properties of matter and energy.

Despite these divisions, there is significant overlap among scientific fields, with chemistry serving as a crucial link.

Interdisciplinary Fields where Chemistry Overlaps:

Biochemistry: - **Relationship:** Overlap between Chemistry and Biology. - **Definition:** The study of the chemical processes occurring in living matter. - **Example:** Chemists working on the isolation, characterization, and biological activities of compounds from medicinal plants.

Geochemistry: - **Relationship:** Overlap between Chemistry and Geology. - **Definition:** The study of the processes that control the abundance, composition, and distribution of chemical compounds and isotopes in geologic environments.

Physical Chemistry / Chemical Physics: - **Relationship:** Overlap between Chemistry and Physics. Both deal with matter and energy, especially at atomic and small molecule properties. - **Physical Chemistry:** The branch of chemistry concerned with the application of the techniques and theories of physics to the study of chemical systems. - **Chemical Physics:** A sub-discipline of chemistry and physics that investigates physicochemical phenomena using techniques from atomic and molecular physics and condensed matter physics. It studies chemical processes from a physics perspective.

Medicinal Chemistry: - **Relationship:** Overlap between Chemistry and Medicine. - **Definition:** A discipline that encloses the design, development, and synthesis of pharmaceutical drugs.

Central Role of Chemistry: At some level, all scientific fields depend on “matter.” Because of this, chemistry has been called the ‘central science’ linking them all together.

📝 Practice Question 12 / የልምምድ ጥያቄ 12 (EUEE-style)

Question / ጥያቄ: Why is chemistry called the “central science”? / ኬሚስትሪ “ማዕከላዊ ሳይንስ” የሚባለው ለምንድን ነው?

Options / አማራጮች:

- A. Because it studies only living things [ሕይወት ያላቸውን ነገሮች ብቻ ስለሚያጠና]
- B. Because at some level all scientific fields depend on matter, and chemistry links them together [በተወሰነ ደረጃ ሁሉም የሳይንስ ዘርፎች በቁስ ላይ ስለሚመሠረቱና ኬሚስትሪ እነሱን ስለሚያገናኝ]
- C. Because it is the only natural science that uses experiments [ሙከራዎችን የሚጠቀም ብቸኛው የተፈጥሮ ሳይንስ ስለሆነ]
- D. Because it studies only rocks and the earth [ድንጋዮችንና ምድርን ብቻ ስለሚያጠና]

Correct Answer / ትክክለኛ መልስ: B

Explanation (English): All scientific fields depend on matter at some level, so chemistry links them together and is called the central science. Option D is the main distractor, since studying rocks and the earth is geology, not the reason for chemistry’s central role.

ማብራሪያ (አማርኛ): ሁሉም የሳይንስ ዘርፎች በተወሰነ ደረጃ በቁስ ላይ ስለሚመሠረቱ ኬሚስትሪ እነሱን ያገናኛል፤ ስለዚህ ማዕከላዊ ሳይንስ ይባላል። አማራጭ D አሳሳች ነው፤ ድንጋዮችንና ምድርን ማጥናት የጂኦሎጂ ሥራ ነው እንጂ የኬሚስትሪ ማዕከላዊነት ምክንያት አይደለም።

📝 Practice Question 13 / የልምምድ ጥያቄ 13 (EUEE-style)

Question / ጥያቄ: Which discipline is the overlap between chemistry and medicine, dealing with the design, development and

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synthesis of pharmaceutical drugs? / በኬሚስትሪና በሕክምና መካከል ያለ፣ የመድኃኒቶችን ንድፍ፣ ልማትና ውህደት የሚመለከት ዘርፍ የትኛው ነው?

Options / አማራጮች:

- A. Geochemistry [ጂኦኬሚስትሪ]
- B. Biochemistry [ባዮኬሚስትሪ]
- C. Chemical physics [ኬሚካል ፊዚክስ]
- D. Medicinal chemistry [መድኃኒታዊ ኬሚስትሪ]

Correct Answer / ትክክለኛ መልስ: D

Explanation (English): Medicinal chemistry is the overlap between chemistry and medicine, covering the design, development and synthesis of pharmaceutical drugs. Biochemistry is the main distractor because it overlaps chemistry with biology, not medicine.

ማብራሪያ (አማርኛ): መድኃኒታዊ ኬሚስትሪ በኬሚስትሪና በሕክምና መካከል ያለ ሲሆን የመድኃኒቶችን ንድፍ፣ ልማትና ውህደት ይመለከታል። ባዮኬሚስትሪ ኬሚስትሪን ከባዮሎጂ ጋር የሚያገናኝ እንጂ ከሕክምና ጋር ስላልሆነ አሳሳች አማራጭ ነው።

📝 Practice Question 14 / የልምምድ ጥያቄ 14 (EUEE-style)

Question / ጥያቄ: Chemists work on the isolation, characterization and biological activities of compounds from medicinal plants. This work is an example of which interdisciplinary field? / ኬሚስቶች ከመድኃኒትነት ካላቸው ዕፅዋት ውህዶችን በመለየት፣ ባህሪያቸውን በመግለጽና ሕይወታዊ እንቅስቃሴያቸውን በማጥናት ይሠራሉ። ይህ ሥራ የየትኛው የጋራ ዘርፍ ምሳሌ ነው?

Options / አማራጮች:

- A. Biochemistry [ባዮኬሚስትሪ]
- B. Geochemistry [ጂኦኬሚስትሪ]
- C. Chemical physics [ኬሚካል ፊዚክስ]
- D. Physical chemistry [ፊዚካል ኬሚስትሪ]

Correct Answer / ትክክለኛ መልስ: A

Explanation (English): The guide gives this work as the example of biochemistry, the overlap between chemistry and biology that studies chemical processes in living matter. Geochemistry is the main distractor because it deals with chemical compounds in geologic environments.

ማብራሪያ (አማርኛ): መመሪያው ይህንን ሥራ የባዮኬሚስትሪ ምሳሌ አድርጎ ይሰጣል፤ ባዮኬሚስትሪ በሕይወት ባለው ቁስ ውስጥ የሚከሰቱ ኬሚካላዊ ሂደቶችን የሚያጠና የኬሚስትሪና የባዮሎጂ መደራረብ ነው። ጂኦኬሚስትሪ በጂኦሎጂካል አካባቢዎች ያሉ ኬሚካላዊ ውህዶችን ስለሚመለከት አሳሳች አማራጭ ነው።

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📝 Practice Question 15 / የልምምድ ጥያቄ 15 (EUEE-style)

Question / ጥያቄ: The study of the processes that control the abundance, composition and distribution of chemical compounds and isotopes in geologic environments is called _____. / በጂኦሎጂካል አካባቢዎች ውስጥ የኬሚካል ውህዶችንና አይሶቶፖችን ብዛት፣ ስብጥርና ስርጭት የሚቆጣጠሩ ሂደቶችን ማጥናት _____ ይባላል።

Options / አማራጮች:

- A. Biochemistry [ባዮኬሚስትሪ]
- B. Chemical physics [ኬሚካል ፊዚክስ]
- C. Geochemistry [ጂኦኬሚስትሪ]
- D. Medicinal chemistry [መድኃኒታዊ ኬሚስትሪ]

Correct Answer / ትክክለኛ መልስ: C

Explanation (English): Geochemistry is the overlap between chemistry and geology and studies these processes in geologic environments. Biochemistry is the main distractor because it studies chemical processes in living matter.

ማብራሪያ (አማርኛ): ጂኦኬሚስትሪ በኬሚስትሪና በጂኦሎጂ መካከል ያለ ዘርፍ ሲሆን እነዚህን ሂደቶች በጂኦሎጂካል አካባቢዎች ያጠናል። ባዮኬሚስትሪ በሕይወት ባለው ቁስ ውስጥ ያሉ ኬሚካላዊ ሂደቶችን ስለሚያጠና አሳሳች አማራጭ ነው።

📝 Practice Question 16 / የልምምድ ጥያቄ 16 (EUEE-style)

Question / ጥያቄ: Which of the following pairs of interdisciplinary field and the sciences it overlaps is **NOT** correct? / ከሚከተሉት የጋራ ዘርፍና የሚደራረባቸው ሳይንሶች ጥንዶች ውስጥ ትክክል ያልሆነው የትኛው ነው?

Options / አማራጮች:

- A. Biochemistry – chemistry and biology [ባዮኬሚስትሪ – ኬሚስትሪና ባዮሎጂ]
- B. Geochemistry – chemistry and geology [ጂኦኬሚስትሪ – ኬሚስትሪና ጂኦሎጂ]
- C. Physical chemistry / chemical physics – chemistry and physics [ፊዚካል ኬሚስትሪ / ኬሚካል ፊዚክስ – ኬሚስትሪና ፊዚክስ]
- D. Medicinal chemistry – chemistry and geology [መድኃኒታዊ ኬሚስትሪ – ኬሚስትሪና ጂኦሎጂ]

Correct Answer / ትክክለኛ መልስ: D

Explanation (English): Medicinal chemistry is the overlap between chemistry and medicine, not geology; the link between chemistry and geology is geochemistry. Options A, B and C match the guide.

ማብራሪያ (አማርኛ): መድኃኒታዊ ኬሚስትሪ በኬሚስትሪና በሕክምና መካከል ያለ እንጂ ከጂኦሎጂ ጋር አይደለም፤ ኬሚስትሪንና ጂኦሎጂን የሚያገናኘው ጂኦኬሚስትሪ ነው። አማራጮች A፣ B እና C ከመመሪያው ጋር ይስማማሉ።

1.3 The Role Chemistry Plays in Production and in the Society

Chemistry is integral to many aspects of our daily lives, contributing significantly to advancements in various sectors and improving societal well-being.

A. Agriculture

Chemistry has revolutionized agriculture, enabling increased food production and crop protection.

Chemical Fertilizers: - Examples: Calcium super phosphate, urea, ammonium sulphate, and sodium nitrate. - Role: Greatly assist in increasing the yield of fruits, vegetables, and other crops.

Pesticides: - Role: Manufactured through chemistry to lessen crop damage. - Types (depending on targeted pest): Fungicides, herbicides, and insecticides. - Impact: Help supply the ever-growing demand for food.

Plastic Pipes for Irrigation: - Role: Chemistry contributes to the manufacturing of better-quality plastic pipes, commonly used in farming. - Impact: Massively increased irrigation, leading to improved growing climates for crops.

📝 Practice Question 17 / የልምምድ ጥያቄ 17 (EUEE-style)

Question / ጥያቄ: Which of the following is an example of a chemical fertilizer? / ከሚከተሉት ውስጥ የኬሚካል ማዳበሪያ ምሳሌ የሆነው የቱ ነው?

Options / አማራጮች:

- A. Penicillin [ፔኒሲሊን]
- B. Urea [ዩሪያ]
- C. Glass [መስታወት]
- D. Cement [ሲሚንቶ]

Correct Answer / ትክክለኛ መልስ: B

Explanation (English): Urea is one of the chemical fertilizers listed in the guide, along with calcium super phosphate, ammonium sulphate and sodium nitrate. Penicillin is the main distractor: it is a medicine used to cure diseases such as pneumonia, not a fertilizer.

ማብራሪያ (አማርኛ): ዩሪያ ከካልሲየም ሱፐር ፎስፌት፣ ከአሞኒየም ሰልፌትና ከሶዲየም ናይትሬት ጋር በመመሪያው ከተዘረዘሩ የኬሚካል ማዳበሪያዎች አንዱ ነው። ፔኒሲሊን እንደ የሳንባ ምች ያሉ በሽታዎችን የሚያድን መድኃኒት እንጂ ማዳበሪያ ስላልሆነ አሳሳች አማራጭ ነው።

📝 Practice Question 18 / የልምምድ ጥያቄ 18 (EUEE-style)

Question / ጥያቄ: Depending on the targeted pest, which of the following are the types of pesticides? / እንደ ዒላማው ተባይ ዓይነት የፀረ-ተባይ ኬሚካሎች ዓይነቶች የትኞቹ ናቸው?

Options / አማራጮች:

- A. Antibiotics, antiseptics and analgesics [አንቲባዮቲኮች፣ አንቲሴፕቲኮች እና የሕመም ማስታገሻዎች]
- B. Fertilizers, preservatives and adulterants [ማዳበሪያዎች፣ ማቆያዎች እና የምግብ ብክለት ንጥረ ነገሮች]
- C. Fungicides, herbicides and insecticides [ፀረ-ፈንገሶች፣ ፀረ-አረሞች እና ፀረ-ነፍሳት]
- D. Disinfectants, tranquilizers and anesthetics [ንጽሕና ማጽጃዎች፣ ማረጋጊያዎች እና ማደንዘዣዎች]

Correct Answer / ትክክለኛ መልስ: C

Explanation (English): The guide lists fungicides, herbicides and insecticides as the types of pesticides, depending on the targeted pest. Option A is the main distractor because those are medicines (drug categories), not pesticides used to lessen crop damage.

ማብራሪያ (አማርኛ): መመሪያው እንደ ዒላማው ተባይ የፀረ-ተባይ ዓይነቶችን ፀረ-ፈንገስ፣ ፀረ-አረምና ፀረ-ነፍሳት ብሎ ይዘረዝራል። አማራጭ A የመድኃኒት ዓይነቶች እንጂ የሰብል ጉዳትን የሚቀንሱ ፀረ-ተባዮች ስላልሆኑ አሳሳች ነው።

📝 Practice Question 19 / የልምምድ ጥያቄ 19 (EUEE-style)

Question / ጥያቄ: What is the impact of better-quality plastic pipes, which chemistry helps to manufacture, in farming? / ኬሚስትሪ ለማምረት የሚረዳቸው የተሻሉ የፕላስቲክ ቧንቧዎች በግብርና ላይ ምን ውጤት አላቸው?

Options / አማራጮች:

- A. Massively increased irrigation, leading to improved growing climates for crops [የመስኖ ሥራ በከፍተኛ ደረጃ መጨመር፣ ለሰብሎች የተሻለ የዕድገት አየር ንብረት መፍጠር]
- B. Preservation of raw meat for a longer period [ጥሬ ሥጋን ለረጅም ጊዜ ማቆየት]
- C. Killing of microbes in toilets, floors and drains [በመጸዳጃ ቤቶች፣ ወለሎችና ቦይዎች ያሉ ረቂቅ ተሕዋስያንን መግደል]
- D. Construction of durable dams and bridges [ዘላቂ ግድቦችንና ድልድዮችን መገንባት]

Correct Answer / ትክክለኛ መልስ: A

Explanation (English): Better-quality plastic pipes, commonly used in farming, massively increased irrigation and improved growing climates for crops. Option D is the main distractor: durable dams and bridges are linked to building construction materials, not plastic irrigation pipes.

ማብራሪያ (አማርኛ): በግብርና ላይ በብዛት የሚያገለግሉ የተሻሉ የፕላስቲክ ቧንቧዎች የመስኖ ሥራን በከፍተኛ ደረጃ ጨምረዋል፤ ለሰብሎችም የተሻለ የዕድገት አየር ንብረት ፈጥረዋል። አማራጭ D አሳሳች ነው፤ ዘላቂ ግድቦችና ድልድዮች ከግንባታ ቁሶች ጋር እንጂ ከመስኖ የፕላስቲክ ቧንቧዎች ጋር አይያያዙም።

📝 Practice Question 20 / የልምምድ ጥያቄ 20 (EUEE-style)

Question / ጥያቄ: Which of the following is the odd one out, that is, **NOT** a chemical fertilizer? / ከሚከተሉት ውስጥ ከሌሎቹ

የተለየው፣ ማለትም የኬሚካል ማዳበሪያ ያልሆነው የትኛው ነው?

Options / አማራጮች:

- A. Calcium super phosphate [ካልሲየም ሱፐር ፎስፌት]
- B. Ammonium sulphate [አሞኒየም ሰልፌት]
- C. Sodium nitrate [ሶዲየም ናይትሬት]
- D. Malathion [ማላቲዮን]

Correct Answer / ትክክለኛ መልስ: D

Explanation (English): Calcium super phosphate, ammonium sulphate and sodium nitrate are listed chemical fertilizers. Malathion is among the pesticides formulated at the Adami Tulu Pesticide Processing Plant, so it is the odd one out. Sodium nitrate is the main distractor: it may not look like a typical fertilizer, but the guide lists it as one.

ማብራሪያ (አማርኛ): ካልሲየም ሱፐር ፎስፌት፣ አሞኒየም ሰልፌትና ሶዲየም ናይትሬት ከተዘረዘሩት የኬሚካል ማዳበሪያዎች ናቸው። ማላቲዮን በአዳሚ ቱሉ የፀረ-ተባይ ማምረቻ ፋብሪካ ከሚዘጋጁ ፀረ-ተባዮች አንዱ ስለሆነ የተለየው እሱ ነው፤ ሶዲየም ናይትሬት የተለመደ ማዳበሪያ ባይመስልም መመሪያው ከማዳበሪያዎች ይዘረዝረዋል፤ ስለዚህ አሳሳች አማራጭ ነው።

B. Food Production

Beyond agriculture, chemistry plays a vital role in processing, preserving, and ensuring the quality of food.

Food Preservatives: - Role: Discovery of different kinds of chemicals that preserve food products for longer periods.

Testing for Adulterants: - Role: Provides methods to test for the presence of adulterants, ensuring the supply of pure foodstuff.

Consumer Benefits: - New technologies derived from chemistry have increased food's: - Availability - Appearance - Nutritional contents - Flavor - Local Example: The preservation of raw meat.

📝 Practice Question 21 / የልምምድ ጥያቄ 21 (EUEE-style)

Question / ጥያቄ: What do we call the chemicals discovered through chemistry that preserve food products for longer periods? / ኬሚስትሪ ባገኛቸው ኬሚካሎች የምግብ ምርቶችን ለረጅም ጊዜ የሚያቆዩት ምን ተብለው ይጠራሉ?

Options / አማራጮች:

- A. Adulterants [የምግብ ብክለት ንጥረ ነገሮች]
- B. Food preservatives [የምግብ ማቆያዎች]
- C. Fertilizers [ማዳበሪያዎች]
- D. Disinfectants [ንጽሕና ማጽጃዎች]

Correct Answer / ትክክለኛ መልስ: B

Explanation (English): Food preservatives are chemicals that keep food products usable for longer periods. Adulterants are the main distractor: chemistry provides tests to detect them so that foodstuff stays pure.

ማብራሪያ (አማርኛ): የምግብ ማቆያዎች የምግብ ምርቶችን ለረጅም ጊዜ የሚያቆዩ ኬሚካሎች ናቸው። ንጹሕ ምግብ እንዲቀርብ ኬሚስትሪ እነሱን የሚለዩ ዘዴዎችን ስለሚሰጥ፣ የምግብ ብክለት ንጥረ ነገሮች (adulterants) አሳሳች አማራጭ ናቸው።

📝 Practice Question 22 / የልምምድ ጥያቄ 22 (EUEE-style)

Question / ጥያቄ: What is the role of the methods chemistry provides to test for the presence of adulterants in food? / ኬሚስትሪ በምግብ ውስጥ ያሉ የምግብ ብክለት ንጥረ ነገሮችን (adulterants) ለመመርመር የሚሰጣቸው ዘዴዎች ምን ሚና አላቸው?

Options / አማራጮች:

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- A. To increase the use of fertilizers [የማዳበሪያ አጠቃቀምን ለመጨመር]
- B. To kill microbes on floors [በወለሎች ላይ ያሉ ረቂቅ ተሕዋስያንን ለመግደል]
- C. To ensure the supply of pure foodstuff [ንጹሕ ምግብ እንዲቀርብ ለማረጋገጥ]
- D. To reduce tension in patients [በታካሚዎች ላይ ያለውን ውጥረት ለመቀነስ]

Correct Answer / ትክክለኛ መልስ: C

Explanation (English): Tests for adulterants help ensure the supply of pure foodstuff. Option B is the main distractor, because killing microbes on floors is the purpose of disinfectants, not of food testing.

ማብራሪያ (አማርኛ): የምግብ ብክለት ንጥረ ነገሮችን የሚመረምሩ ዘዴዎች ንጹሕ ምግብ እንዲቀርብ ለማረጋገጥ ይረዳሉ። አማራጭ B አሳሳች ነው፤ በወለሎች ላይ ያሉ ረቂቅ ተሕዋስያንን መግደል የንጽሕና ማጽጃዎች (disinfectants) ዓላማ እንጂ የምግብ ምርመራ አይደለም።

📝 Practice Question 23 / የልምምድ ጥያቄ 23 (EUEE-style)

Question / ጥያቄ: New technologies derived from chemistry have improved food in many ways. Which of the following is **NOT** one of these benefits? / ከኬሚስትሪ የተገኙ አዳዲስ ቴክኖሎጂዎች ምግብን በብዙ መንገድ አሻሽለዋል። ከሚከተሉት ውስጥ ከእነዚህ ጥቅሞች አንዱ ያልሆነው የትኛው ነው?

Options / አማራጮች:

- A. Higher risk of adulteration [የምግብ የመበከል አደጋ መጨመር]
- B. Availability [የምግብ ተገኝነት]
- C. Nutritional contents [የምግብ ንጥረ ነገር ይዘት]
- D. Flavor [ጣዕም]

Correct Answer / ትክክለኛ መልስ: A

Explanation (English): The guide says new technologies from chemistry increased the availability, appearance, nutritional contents and flavor of food. A higher risk of adulteration is not a benefit; chemistry provides tests for adulterants so that food stays pure. The main distractor is availability (option B), which is a real benefit.

ማብራሪያ (አማርኛ): መመሪያው ከኬሚስትሪ የተገኙ አዳዲስ ቴክኖሎጂዎች የምግብን ተገኝነት፣ መልክ፣ የንጥረ ነገር ይዘትና ጣዕም እንዳሻሻሉ ይገልጻል። የመበከል አደጋ መጨመር ጥቅም አይደለም፤ ኬሚስትሪ ምግብ ንጹሕ እንዲሆን የብክለት ንጥረ ነገሮችን የሚመረምሩ ዘዴዎችን ይሰጣል። ዋናው አሳሳች አማራጭ ተገኝነት (B) ነው፤ ይህ እውነተኛ ጥቅም ነው።

C. Medicine

Chemistry is a cornerstone of modern medicine, providing life-saving drugs and treatment methods.

Life-Saving Medicines: - Early Discoveries: Sulphur drugs and penicillin led to cures for dysentery and pneumonia. - Cancer Therapy: Cisplatin and Taxol are effective for cancer treatment. - HIV/AIDS Treatment: AZT fights the multiplication of the HIV virus, prolonging the life of victims, though it does not cure the disease.

Common Drug Categories Provided by Chemistry:

Disinfectants: - Purpose: Used to kill microbes present in toilets, floors, and drains. - Example: Sanitizers used for Covid-19.

Analgesics (Painkillers): - Purpose: Drugs used to achieve analgesia, which is relief from pain.

Anesthetics: - Purpose: Relieve pain, making medical operations more effective.

Antibiotics: - Purpose: Used to control infection and cure diseases.

Antiseptics: - Purpose: Used to prevent contamination of wounds by bacteria.

Tranquilizers: - Purpose: Reduce tension and bring about calm and peace for patients suffering from mental diseases.

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📝 Practice Question 24 / የልምምድ ጥያቄ 24 (EUEE-style)

Question / ጥያቄ: Which statement describes analgesics? / የትኛው መግለጫ የሕመም ማስታገሻዎችን (analgesics) ይገልጻል?

Options / አማራጮች:

- A. They prevent contamination of wounds by bacteria [የቁስል በባክቴሪያ መበከልን ይከላከላሉ]
- B. They reduce tension and bring calm to patients with mental diseases [በአእምሮ ሕመም የሚሠቃዩ ታካሚዎችን ውጥረት ይቀንሳሉ፣ መረጋጋትም ያመጣሉ]
- C. They are drugs used to achieve relief from pain [ከሕመም እፎይታ ለማግኘት የሚያገለግሉ መድኃኒቶች ናቸው]
- D. They are used to control infection [ኢንፌክሽንን ለመቆጣጠር ያገለግላሉ]

Correct Answer / ትክክለኛ መልስ: C

Explanation (English): Analgesics, or painkillers, are drugs used to achieve analgesia, which is relief from pain. Option D is the main distractor, because controlling infection is the purpose of antibiotics.

ማብራሪያ (አማርኛ): የሕመም ማስታገሻዎች (analgesics) ከሕመም እፎይታ ለማግኘት የሚያገለግሉ መድኃኒቶች ናቸው። አማራጭ D አሳሳች ነው፤ ኢንፌክሽንን መቆጣጠር የአንቲባዮቲኮች ዓላማ ነው።

📝 Practice Question 25 / የልምምድ ጥያቄ 25 (EUEE-style)

Question / ጥያቄ: Which drug category is used to prevent contamination of wounds by bacteria? / የቁስልን በባክቴሪያ መበከል ለመከላከል የሚያገለግለው የመድኃኒት ምድብ የትኛው ነው?

Options / አማራጮች:

- A. Analgesics [የሕመም ማስታገሻዎች]
- B. Tranquilizers [ማረጋጊያዎች]
- C. Disinfectants [ንጽሕና ማጽጃዎች]
- D. Antiseptics [አንቲሴፕቲኮች]

Correct Answer / ትክክለኛ መልስ: D

Explanation (English): Antiseptics are used to prevent contamination of wounds by bacteria. Disinfectants are the main distractor, since they are used to kill microbes present in toilets, floors and drains.

ማብራሪያ (አማርኛ): አንቲሴፕቲኮች የቁስልን በባክቴሪያ መበከል ለመከላከል ያገለግላሉ። ንጽሕና ማጽጃዎች (disinfectants) በመጸዳጃ ቤቶች፣ ወለሎችና ቦይዎች ያሉ ረቂቅ ተሕዋስያንን ለመግደል ስለሚያገለግሉ አሳሳች አማራጭ ናቸው።

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📝 Practice Question 26 / የልምምድ ጥያቄ 26 (EUEE-style)

Question / ጥያቄ: A patient suffering from a mental disease needs a drug that reduces tension and brings calm and peace. Which drug category is suitable? / በአእምሮ ሕመም የሚሠቃይ ታካሚ ውጥረትን የሚቀንስና መረጋጋትና ሰላም የሚያመጣ መድኃኒት ይፈልጋል። የትኛው የመድኃኒት ምድብ ተስማሚ ነው?

Options / አማራጮች:

- A. Anesthetics [ማደንዘዣዎች]
- B. Tranquilizers [ማረጋጊያዎች]
- C. Antibiotics [አንቲባዮቲኮች]
- D. Analgesics [የሕመም ማስታገሻዎች]

Correct Answer / ትክክለኛ መልስ: B

Explanation (English): Tranquilizers reduce tension and bring calm and peace to patients suffering from mental diseases.

Anesthetics are the main distractor, because they relieve pain to make medical operations more effective, not to calm the mind.

ማብራሪያ (አማርኛ): ማረጋጊያዎች በአእምሮ ሕመም የሚሠቃዩ ታካሚዎችን ውጥረት ይቀንሳሉ፣ መረጋጋትና ሰላምም ያመጣሉ። ማደንዘዣዎች የሕክምና ሥራዎች ይበልጥ ውጤታማ እንዲሆኑ ሕመምን የሚያስታግሱ እንጂ አእምሮን የሚያረጋጉ ስላልሆኑ አሳሳች አማራጭ ናቸው።

📝 Practice Question 27 / የልምምድ ጥያቄ 27 (EUEE-style)

Question / ጥያቄ: Sanitizers used for Covid-19, which kill microbes present in toilets, floors and drains, belong to which category? / ለኮቪድ-19 የሚያገለግሉ ሳኒታይዘሮች፣ በመጸዳጃ ቤቶች፣ ወለሎችና ቦይዎች ያሉ ረቂቅ ተሕዋስያንን የሚገድሉ፣ በየትኛው ምድብ ይካተታሉ?

Options / አማራጮች:

- A. Disinfectants [ንጽሕና ማጽጃዎች]
- B. Antiseptics [አንቲሴፕቲኮች]
- C. Antibiotics [አንቲባዮቲኮች]
- D. Anesthetics [ማደንዘዣዎች]

Correct Answer / ትክክለኛ መልስ: A

Explanation (English): The guide places sanitizers used for Covid-19 under disinfectants, which kill microbes in toilets, floors and drains. Antiseptics are the main distractor, because they are used to prevent bacterial contamination of wounds.

ማብራሪያ (አማርኛ): መመሪያው ለኮቪድ-19 የሚያገለግሉ ሳኒታይዘሮችን በንጽሕና ማጽጃዎች (disinfectants) ሥር ያስቀምጣል፤ እነዚህም በመጸዳጃ ቤቶች፣ ወለሎችና ቦይዎች ያሉ ረቂቅ ተሕዋስያንን ይገድላሉ። አንቲሴፕቲኮች የቁስልን የባክቴሪያ መበከል ለመከላከል ስለሚያገለግሉ አሳሳች አማራጭ ናቸው።

📝 Practice Question 28 / የልምምድ ጥያቄ 28 (EUEE-style)

Question / ጥያቄ: Which statement about life-saving medicines is correct? / ስለሕይወት አድን መድኃኒቶች የትኛው መግለጫ ትክክል ነው?

Options / አማራጮች:

- A. AZT cures HIV/AIDS completely [AZT ኤችአይቪ/ኤድስን ሙሉ በሙሉ ይፈውሳል]
- B. Cisplatin and Taxol cure dysentery and pneumonia [ሲስፕላቲንና ታክሶል ተቅማጥንና የሳንባ ምችን ይፈውሳሉ]
- C. AZT fights the multiplication of the HIV virus and prolongs the life of victims, though it does not cure the disease [AZT የኤችአይቪ ቫይረስ መባዛትን ይዋጋል፣ የተጠቂዎችንም ሕይወት ያራዝማል፤ በሽታውን ግን አይፈውስም]
- D. Penicillin is used for cancer therapy [ፔኒሲሊን ለካንሰር ሕክምና ያገለግላል]

Correct Answer / ትክክለኛ መልስ: C

Explanation (English): AZT fights the multiplication of the HIV virus and prolongs life, but it does not cure the disease. Option B is the main distractor: it swaps the uses, since sulphur drugs and penicillin led to cures for dysentery and pneumonia, while cisplatin and Taxol are used for cancer treatment.

ማብራሪያ (አማርኛ): AZT የኤችአይቪ ቫይረስ መባዛትን ይዋጋል፣ ሕይወትንም ያራዝማል፤ በሽታውን ግን አይፈውስም። አማራጭ B አሳሳች ነው፤ አጠቃቀሞቹን ይቀያይራል፤ ምክንያቱም የሰልፈር መድኃኒቶችና ፔኒሲሊን ለተቅማጥና ለሳንባ ምች ፈውስ ሲሆኑ ሲስፕላቲንና ታክሶል ለካንሰር ሕክምና ያገለግላሉ።

D. Building Construction Materials

Chemistry provides essential materials that enable the construction of modern, durable, and safe infrastructure.

Building Resources: - Chemistry contributes resources such as glass, steel, and cement. - Impact: Helps in the construction of

safer houses, multi-story structures, and longlasting, durable dams and bridges. - **Example: The Grand Ethiopian Renaissance Dam (GERD):** - A 6,450 MW hydropower project nearing completion on the Blue Nile in Ethiopia. - Located about 30 km upstream of the border with Sudan. - Set to be the largest hydropower project in Africa, showcasing the role of chemistry in large-scale engineering.

📝 Practice Question 29 / የልምምድ ጥያቄ 29 (EUEE-style)

Question / ጥያቄ: Which of the following are building resources to which chemistry contributes? / ኬሚስትሪ አስተዋጽኦ ከሚያደርግባቸው የግንባታ ሀብቶች ውስጥ የትኞቹ ናቸው?

Options / አማራጮች:

- A. Glass, steel and cement [መስታወት፣ ብረት እና ሲሚንቶ]
- B. Urea, ammonium sulphate and sodium nitrate [ዩሪያ፣ አሞኒየም ሰልፌት እና ሶዲየም ናይትሬት]
- C. Cisplatin, Taxol and AZT [ሲስፕላቲን፣ ታክሶል እና AZT]
- D. Fungicides, herbicides and insecticides [ፀረ-ፈንገሶች፣ ፀረ-አረሞች እና ፀረ-ነፍሳት]

Correct Answer / ትክክለኛ መልስ: A

Explanation (English): The guide names glass, steel and cement as building resources that chemistry contributes. Option B is the main distractor, since those are chemical fertilizers used in agriculture.

ማብራሪያ (አማርኛ): መመሪያው ኬሚስትሪ የሚያበረክታቸው የግንባታ ሀብቶች መስታወት፣ ብረትና ሲሚንቶ እንደሆኑ ይገልጻል። አማራጭ B በግብርና የሚያገለግሉ የኬሚካል ማዳበሪያዎች ስለሆኑ አሳሳች አማራጭ ነው።

📝 Practice Question 30 / የልምምድ ጥያቄ 30 (EUEE-style)

Question / ጥያቄ: Which statement about the Grand Ethiopian Renaissance Dam (GERD) is correct according to the guide? / በመመሪያው መሠረት ስለ ታላቁ የኢትዮጵያ ህዳሴ ግድብ (GERD) የትኛው መግለጫ ትክክል ነው?

Options / አማራጮች:

- A. It is a 6,450 MW thermal power plant in Addis Ababa [በአዲስ አበባ የሚገኝ 6,450 ሜጋ ዋት የሙቀት ኃይል ማመንጫ ነው]
- B. It is a 6,450 MW hydropower project on the Blue Nile [በብሉ ናይል ላይ የሚገኝ 6,450 ሜጋ ዋት የውሃ ኃይል ፕሮጀክት ነው]
- C. It is a 450 MW hydropower project on the Blue Nile [በብሉ ናይል ላይ የሚገኝ 450 ሜጋ ዋት የውሃ ኃይል ፕሮጀክት ነው]
- D. It is a 6,450 MW hydropower project on the Awash River [በአዋሽ ወንዝ ላይ የሚገኝ 6,450 ሜጋ ዋት የውሃ ኃይል ፕሮጀክት ነው]

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Correct Answer / ትክክለኛ መልስ: B

Explanation (English): GERD is a 6,450 MW hydropower project on the Blue Nile, located about 30 km upstream of the border with Sudan. Option D is the main distractor because it gives the right capacity but the wrong river.

ማብራሪያ (አማርኛ): ህዳሴ ግድብ በብሉ ናይል ላይ የሚገኝ 6,450 ሜጋ ዋት የውሃ ኃይል ፕሮጀክት ሲሆን ከሱዳን ድንበር 30 ኪ.ሜ ያህል ወደ ላይ ይገኛል። አማራጭ D ትክክለኛውን አቅም ቢሰጥም ወንዙን ስለሚሳሳት አሳሳች አማራጭ ነው።

📝 Practice Question 31 / የልምምድ ጥያቄ 31 (EUEE-style)

Question / ጥያቄ: Which of the following is **NOT** an impact of the building resources that chemistry provides? / ኬሚስትሪ ከሚያቀርባቸው የግንባታ ሀብቶች ውጤቶች ውስጥ ያልሆነው የትኛው ነው?

Options / አማራጮች:

- A. Safer houses [ይበልጥ ደኅንነታቸው የተጠበቀ ቤቶች]

- B. Multi-story structures [ባለ ብዙ ፎቅ ሕንፃዎች]
- C. Long-lasting, durable dams and bridges [ረጅም ዘመን የሚቆዩ ዘላቂ ግድቦችና ድልድዮች]
- D. Increased yield of fruits and vegetables [የፍራፍሬና የአትክልት ምርት መጨመር]

Correct Answer / ትክክለኛ መልስ: D

Explanation (English): Glass, steel and cement help build safer houses, multi-story structures and durable dams and bridges. Increased yield of fruits and vegetables is the result of chemical fertilizers in agriculture, so it is the answer.

ማብራሪያ (አማርኛ): መስታውት፣ ብረትና ሲሚንቶ ደኅንነቱ የተጠበቀ ቤት፣ ባለ ብዙ ፎቅ ሕንፃዎችና ዘላቂ ግድቦችና ድልድዮች ለመገንባት ይረዳሉ። የፍራፍሬና የአትክልት ምርት መጨመር የግብርና ኬሚካል ማዳበሪያዎች ውጤት ስለሆነ መልሱ D ነው።

Summary of Societal Role: Chemistry plays a significant role in the advancement and growth of several industries. However, it is also crucial to manage the production of potentially toxic substances that can negatively affect the environment and lead to issues like climate change.

1.4 Some Common Chemical Industries in Ethiopia

Chemical industries are a vital part of the economy, transforming natural resources into a wide array of products essential for modern life.

Industry: An economic activity concerned with the processing of raw materials and the manufacture of goods in factories. - Alternatively, it can be interpreted as a group of companies linked by their primary business activities, categorized by their largest sources of revenue.

Government Focus: The Ethiopian government has actively expanded industries, including constructing several industrial parks, over the past two decades.

1.4.1 What Chemical Industries Comprise

Chemical industries are diverse and central to the world economy, converting natural resources into various products.

Core Manufacturing Areas: - Inorganic- and organic-industrial chemicals - Explosives - Fragrances - Agrochemicals - Polymers and rubber - Ceramic products - Petrochemicals - Oleochemicals (oils, fats, and waxes) - Flavors

Raw Materials: Convert natural resources such as oil, natural gas, air, water, metals, and minerals.

1.4.2 Categorization of Chemical Industries

Chemical industries are further categorized based on their products and processes: - Industrial inorganic chemicals - Plastics, materials, and synthetics - Drugs - Soap, cleaners, and toilet goods - Paints and allied products - Industrial organic chemicals - Agricultural chemicals - Miscellaneous chemical products

1.4.3 Chemical Products

Chemical Products: Products manufactured, processed, sold, or distributed by a company that are chemical substances or that contain chemical substances.

There are three general classes of chemical products: 1. **Basic chemicals:** - Examples: Alkalis, acids, organic chemicals, and salts. 2. **Chemical products to be used in further manufactures:** - Examples: Plastic materials, synthetic fibers, pigments, and dry colors. 3. **Finished chemical products:** - Purpose: Used for ultimate consumption or as materials/supplies in other industries. - Examples: Cosmetics, drugs, soaps, fertilizers, paints, and explosives.

📝 Practice Question 32 / የልምምድ ጥያቄ 32 (EUEE-style)

Question / ጥያቄ: Which of the following best defines *industry*? / ከሚከተሉት ውስጥ ኢንዱስትሪን በተሻለ ሁኔታ የሚተረጉመው የቱ

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ነው?

Options / አማራጮች:

- A. An economic activity concerned only with growing crops [ሰብል በማምረት ብቻ የተወሰነ ኢኮኖሚያዊ እንቅስቃሴ]
- B. A place where only laboratory experiments are done [የላቦራቶሪ ሙከራዎች ብቻ የሚደረጉበት ቦታ]
- C. An economic activity concerned with the processing of raw materials and the manufacture of goods in factories [ጥሬ ዕቃዎችን በማቀነባበርና በፋብሪካዎች ሸቀጦችን በማምረት ላይ የተሰማራ ኢኮኖሚያዊ እንቅስቃሴ]
- D. The study of the properties of substances [የንጥረ ነገሮችን ባህሪያት ማጥናት]

Correct Answer / ትክክለኛ መልስ: C

Explanation (English): Industry is an economic activity concerned with processing raw materials and manufacturing goods in factories. Option D is the main distractor, because studying the properties of substances is part of the definition of chemistry, not of industry.

ማብራሪያ (አማርኛ): ኢንዱስትሪ ጥሬ ዕቃዎችን በማቀነባበርና በፋብሪካዎች ሸቀጦችን በማምረት ላይ የተሰማራ ኢኮኖሚያዊ እንቅስቃሴ ነው። አማራጭ D አሳሳች ነው፤ የንጥረ ነገሮችን ባህሪያት ማጥናት የኬሚስትሪ ትርጉም አካል እንጂ የኢንዱስትሪ አይደለም።

📝 Practice Question 33 / የልምምድ ጥያቄ 33 (EUEE-style)

Question / ጥያቄ: Plastic materials, synthetic fibers and pigments belong to which general class of chemical products? / የፕላስቲክ ቁሶች፣ ሰው ሠራሽ ፋይበሮችና ቀለማት (pigments) በየትኛው አጠቃላይ የኬሚካል ምርቶች ክፍል ይካተታሉ?

Options / አማራጮች:

- A. Basic chemicals [መሠረታዊ ኬሚካሎች]
- B. Chemical products to be used in further manufactures [ለተጨማሪ ምርት የሚያገለግሉ የኬሚካል ምርቶች]
- C. Finished chemical products [የተጠናቀቁ የኬሚካል ምርቶች]
- D. Raw materials such as oil and natural gas [እንደ ነዳጅ ዘይትና የተፈጥሮ ጋዝ ያሉ ጥሬ ዕቃዎች]

Correct Answer / ትክክለኛ መልስ: B

Explanation (English): The guide lists plastic materials, synthetic fibers, pigments and dry colors as chemical products used in further manufactures. Finished chemical products is the main distractor: those are for ultimate consumption, like cosmetics, drugs and soaps.

ማብራሪያ (አማርኛ): መመሪያው የፕላስቲክ ቁሶችን፣ ሰው ሠራሽ ፋይበሮችን፣ ቀለማትንና ደረቅ ቀለሞችን ለተጨማሪ ምርት የሚያገለግሉ የኬሚካል ምርቶች አድርጎ ይዘረዝራል። የተጠናቀቁ የኬሚካል ምርቶች ለመጨረሻ ፍጆታ የሚውሉ (እንደ ኮስሞቲክስ፣ መድኃኒትና ሳሙና) ስለሆኑ አሳሳች አማራጭ ናቸው።

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📝 Practice Question 34 / የልምምድ ጥያቄ 34 (EUEE-style)

Question / ጥያቄ: Which option correctly matches an example to each of the three general classes of chemical products (basic chemicals; products for further manufactures; finished products)? / የሦስቱን አጠቃላይ የኬሚካል ምርቶች ክፍሎች (መሠረታዊ ኬሚካሎች፤ ለተጨማሪ ምርት የሚያገለግሉ ምርቶች፤ የተጠናቀቁ ምርቶች) እያንዳንዳቸውን ከምሳሌ ጋር በትክክል ያዛመደው የትኛው አማራጭ ነው?

Options / አማራጮች:

- A. Basic: cosmetics; further manufactures: alkalis; finished: pigments [መሠረታዊ፦ ኮስሞቲክስ፤ ለተጨማሪ ምርት፦ አልካላይዎች፤ የተጠናቀቀ፦ ቀለማት]
- B. Basic: salts; further manufactures: soaps; finished: acids [መሠረታዊ፦ ጨዎች፤ ለተጨማሪ ምርት፦ ሳሙናዎች፤ የተጠናቀቀ፦ አሲዶች]
- C. Basic: acids; further manufactures: drugs; finished: plastic materials [መሠረታዊ፦ አሲዶች፤ ለተጨማሪ ምርት፦ መድኃኒቶች፤ የተጠናቀቀ፦ የፕላስቲክ ቁሶች]

- D. Basic: alkalis; further manufactures: synthetic fibers; finished: fertilizers [*መሠረታዊ፦ አልካላይዎች፤ ለተጨማሪ ምርት፦ ሰው ሠራሽ ፋይበሮች፤ የተጠናቀቀ፦ ማዳበሪያዎች*]

Correct Answer / ትክክለኛ መልስ: D

Explanation (English): Alkalis are basic chemicals, synthetic fibers are used in further manufactures, and fertilizers are finished chemical products. Option C is the main distractor: acids are basic chemicals, but drugs are finished products and plastic materials are used in further manufactures, so its last two matches are reversed.

ማብራሪያ (አማርኛ): አልካላይዎች መሠረታዊ ኬሚካሎች ናቸው፣ ሰው ሠራሽ ፋይበሮች ለተጨማሪ ምርት ያገለግላሉ፣ ማዳበሪያዎች ደግሞ የተጠናቀቁ የኬሚካል ምርቶች ናቸው። አማራጭ C አሳሳች ነው፤ አሲዶች መሠረታዊ ቢሆኑም መድኃኒቶች የተጠናቀቁ ምርቶች ሲሆኑ የፕላስቲክ ቁሶች ለተጨማሪ ምርት የሚያገለግሉ ናቸው፤ ስለዚህ የመጨረሻዎቹ ሁለት ተዛማጆቹ ተገላብጠዋል።

📝 Practice Question 35 / የልምምድ ጥያቄ 35 (EUEE-style)

Question / ጥያቄ: Which of the following is a finished chemical product, used for ultimate consumption? / *ከሚከተሉት ውስጥ ለመጨረሻ ፍጆታ የሚውል የተጠናቀቀ የኬሚካል ምርት የሆነው የትኛው ነው?*

Options / አማራጮች:

- A. Alkalis [*አልካላይዎች*]
- B. Plastic materials [*የፕላስቲክ ቁሶች*]
- C. Drugs [*መድኃኒቶች*]
- D. Synthetic fibers [*ሰው ሠራሽ ፋይበሮች*]

Correct Answer / ትክክለኛ መልስ: C

Explanation (English): Drugs are listed as finished chemical products, together with cosmetics, soaps, fertilizers, paints and explosives. Plastic materials and synthetic fibers are the main distractors, since they are used in further manufactures.

ማብራሪያ (አማርኛ): መድኃኒቶች ከኮስሞቲክስ፣ ከሳሙና፣ ከማዳበሪያ፣ ከቀለምና ከፈንጂ ጋር የተጠናቀቁ የኬሚካል ምርቶች ሆነው ተዘርዝረዋል። የፕላስቲክ ቁሶችና ሰው ሠራሽ ፋይበሮች ለተጨማሪ ምርት ስለሚያገለግሉ አሳሳች አማራጮች ናቸው።

1.4.4 Chemical Industries and Products in Ethiopia

Ethiopia has a growing number of medium and large-scale chemical and chemical product industries.

Key Chemicals Produced in Ethiopia: - Aluminum sulphate - Caustic soda (Sodium hydroxide, NaOH) - Soda ash (Trisodium hydrogen dicarbonate dihydrate, $Na_3H(CO_3)_2{\cdot}2H_2O$, also known as Trona) - Carbon dioxide - Bleaching chemicals - Magnesium oxide (MgO) - Pesticides

Key Chemical Products Manufactured in Ethiopia: - Soap and detergent - Cement - Paints - Building materials - Cosmetics - Plastic - Natural gum - Candle - Glass - Sugar - Tyre - Pulp and paper - Pharmaceuticals - Tobacco

Examples of Large and Medium Scale Chemical Enterprises in Ethiopia:

The following lists specific enterprises, their locations, and their primary products: - **Chorra Gas & Chemical products (Addis Ababa):** Plastic, chemicals, petroleum products, Aluminum sulphate, sulphuric acid. - **Ziway Caustic Soda Factory (Ziway):** Sodium hydroxide. - **Abijata Soda Ash Factory (Bulbula):** Trona ($Na_3H(CO_3)_2{\cdot}2H_2O$). - **Repi Soap & Detergent P.L.C (Addis Ababa):** Soap and detergent. - **Adola Magnesium Oxide Factory (Adolla):** Magnesium oxide. - **Adami Tulu Pesticide Processing Plant (Adami-Tulu):** Formulates malathion, endosulfan, diazinon, fenitrothion, and dimethoate. - **Nefas Silk Paints Factory (Addis Ababa):** Paints, varnishes, antirusts, and glues. - **Modern Building Industries (Addis Ababa):** Cement and cement products, ceramics, paints, sanitary ware, adhesives, glues, plastic rubber, terrazzo tiles, cultured marble. - **Kadisco Chemical Industry (Addis Ababa):** Paints, coatings, and adhesives. - **Tadesse Filatea PLC (Woliso):** Soap, detergent, corrugated iron, nail, infant milk formula. - **Etab Laundry Soap Factory (Hawassa):** Soap and detergent. - **Get-Eshet Detergent Manufacturing and Packing P.L.C (Bishoftu):** Detergent products and Leather chemical inputs. -

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Ethio-Asia Industries S.C (Addis Ababa): Soap and detergent. - **Y.B Cosmetics (Sheger City):** Cosmetics and perfume. - **Mekab PLC (Cosmetics) (Addis Ababa):** Hair oil, shampoo, conditioner, body oil, vaseline, body lotion, detergents, and plastic mouldings. - **BEKAS Chemicals PLC (Addis Ababa):** Detergents, cosmetic products, plastic packing materials, industrial surfactants, and putty. - **Arbaminch Textile Share Company (Arbaminch):** Textile and fabric products. - **Teamco Soap Factory (Burayu):** Soap and detergent.

Other Significant Chemical Product Industries in Ethiopia: - **Cement:** Mugher, Diredawa, Mesobo, Derba, Midroc, Dangote. - **Sugar:** Metehara, Wonji, Finchaa, Omokuraz. - **Paper and pulp:** Wonji. - **Pharmaceuticals:** Addis, Ethiopia, Adigrat. - **Tyre:** Horizon Addis Tyre.

📝 Practice Question 36 / የልምምድ ጥያቄ 36 (EUEE-style)

Question / ጥያቄ: Which chemical is produced by the Ziway Caustic Soda Factory? / የዝዋይ ካስቲክ ሶዳ ፋብሪካ የሚያመርተው ኬሚካል የትኛው ነው?

Options / አማራጮች:

- A. Magnesium oxide [ማግኒዥየም ኦክሳይድ]
- B. Sodium hydroxide (NaOH) [ሶዲየም ሃይድሮክሳይድ (NaOH)]
- C. Aluminum sulphate [አሉሚኒየም ሰልፌት]
- D. Soda ash [ሶዳ አሽ]

Correct Answer / ትክክለኛ መልስ: B

Explanation (English): Ziway Caustic Soda Factory in Ziway produces sodium hydroxide, which is caustic soda (NaOH). Soda ash is the main distractor: it is produced by the Abijata Soda Ash Factory, not at Ziway.

ማብራሪያ (አማርኛ): በዝዋይ የሚገኘው የዝዋይ ካስቲክ ሶዳ ፋብሪካ ሶዲየም ሃይድሮክሳይድ (ካስቲክ ሶዳ፣ NaOH) ያመርታል። ሶዳ አሽ በአቢጃታ ሶዳ አሽ ፋብሪካ የሚመረት እንጂ በዝዋይ ስላልሆነ አሳሳች አማራጭ ነው።

📝 Practice Question 37 / የልምምድ ጥያቄ 37 (EUEE-style)

Question / ጥያቄ: Where is the Abijata Soda Ash Factory located? / የአቢጃታ ሶዳ አሽ ፋብሪካ የሚገኘው የት ነው?

Options / አማራጮች:

- A. Ziway [ዝዋይ]
- B. Adolla [አዶላ]
- C. Bulbula [ቡልቡላ]
- D. Woliso [ወሊሶ]

Correct Answer / ትክክለኛ መልስ: C

Explanation (English): The guide places the Abijata Soda Ash Factory in Bulbula, where it produces Trona. Ziway is the main distractor because it is the location of the Ziway Caustic Soda Factory, which makes sodium hydroxide.

ማብራሪያ (አማርኛ): መመሪያው የአቢጃታ ሶዳ አሽ ፋብሪካን በቡልቡላ ያስቀምጣል፤ እዚያም ትሮናን ያመርታል። ዝዋይ ሶዲየም ሃይድሮክሳይድ የሚያመርተው የዝዋይ ካስቲክ ሶዳ ፋብሪካ መገኛ ስለሆነ አሳሳች አማራጭ ነው።

📝 Practice Question 38 / የልምምድ ጥያቄ 38 (EUEE-style)

Question / ጥያቄ: Which enterprise is correctly matched with its location? / ከሚከተሉት ውስጥ ድርጅቱን ከመገኛው ጋር በትክክል ያዛመደው የትኛው ነው?

Options / አማራጮች:

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- A. Etab Laundry Soap Factory – Hawassa [ኤታብ የልብስ ሳሙና ፋብሪካ – ሐዋሳ]
- B. Tadesse Filatea PLC – Bishoftu [ታደሰ ፊላቴአ ኃ.የተ.የግ.ማ – ቢሾፍቱ]
- C. Teamco Soap Factory – Woliso [ቲምኮ ሳሙና ፋብሪካ – ወሊሶ]
- D. Get-Eshet Detergent Manufacturing and Packing P.L.C – Burayu [ጌት-እሸት ዲተርጀንት ማምረቻና ማሸጊያ ኃ.የተ.የግ.ማ – ቡራዩ]

Correct Answer / ትክክለኛ መልስ: A

Explanation (English): Etab Laundry Soap Factory is in Hawassa. The other options swap locations: Tadesse Filatea is in Woliso, Teamco Soap Factory is in Burayu, and Get-Eshet is in Bishoftu.

ማብራሪያ (አማርኛ): ኤታብ የልብስ ሳሙና ፋብሪካ የሚገኘው በሐዋሳ ነው። ሌሎቹ አማራጮች መገኛዎችን ይቀያይራሉ፤ ታደሰ ፊላቴአ በወሊሶ፣ ቲምኮ ሳሙና ፋብሪካ በቡራዩ፣ ጌት-እሸት ደግሞ በቢሾፍቱ ይገኛሉ።

📝 Practice Question 39 / የልምምድ ጥያቄ 39 (EUEE-style)

Question / ጥያቄ: Which chemical produced in Ethiopia has the formula $Na_3H(CO_3)_2{\cdot}2H_2O$ and is also known as Trona? / በኢትዮጵያ ከሚመረቱ ኬሚካሎች ውስጥ ቀመሩ $Na_3H(CO_3)_2{\cdot}2H_2O$ ሆኖ ትሮና ተብሎም የሚታወቀው የትኛው ነው?

Options / አማራጮች:

- A. Caustic soda [ካስቲክ ሶዳ]
- B. Magnesium oxide [ማግኒዥየም ኦክሳይድ]
- C. Aluminum sulphate [አሉሚኒየም ሰልፌት]
- D. Soda ash [ሶዳ አሽ]

Correct Answer / ትክክለኛ መልስ: D

Explanation (English): Soda ash is trisodium hydrogen dicarbonate dihydrate, $Na_3H(CO_3)_2{\cdot}2H_2O$, also known as Trona. Caustic soda is the main distractor, since it is sodium hydroxide with the formula NaOH.

ማብራሪያ (አማርኛ): ሶዳ አሽ ትራይሶዲየም ሃይድሮጂን ዳይካርቦኔት ዳይሃይድሬት ሲሆን ቀመሩ $Na_3H(CO_3)_2{\cdot}2H_2O$ ነው፤ ትሮና ተብሎም ይታወቃል። ካስቲክ ሶዳ ሶዲየም ሃይድሮክሳይድ ሆኖ ቀመሩ NaOH ስለሆነ አሳሳች አማራጭ ነው።

📝 Practice Question 40 / የልምምድ ጥያቄ 40 (EUEE-style)

Question / ጥያቄ: Which of the following is **NOT** listed as a cement industry in Ethiopia? / ከሚከተሉት ውስጥ በኢትዮጵያ ከሲሚንቶ ኢንዱስትሪዎች ተርታ ያልተዘረዘረው የትኛው ነው?

Options / አማራጮች:

- A. Mugher [ሙገር]
- B. Dangote [ዳንጎቴ]
- C. Derba [ደርባ]
- D. Metehara [መተሐራ]

Correct Answer / ትክክለኛ መልስ: D

Explanation (English): Mugher, Dangote and Derba are listed among the cement industries, while Metehara is listed among the sugar industries. Dangote is the main distractor: it may be less familiar than the others, but the guide includes it in the cement list.

ማብራሪያ (አማርኛ): ሙገር፣ ዳንጎቴና ደርባ ከሲሚንቶ ኢንዱስትሪዎች ተርታ ተዘርዝረዋል፤ መተሐራ ግን ከስኳር ኢንዱስትሪዎች ተርታ ተዘርዝሯል። ዳንጎቴ ከሌሎቹ ያነሰ የታወቀ ሊመስል ስለሚችል አሳሳች አማራጭ ነው፤ መመሪያው ግን በሲሚንቶ ዝርዝር ውስጥ ያካትተዋል።

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Key Terms of the Unit

- **Analgesics:** Drugs used to relieve pain.
- **Antibiotics:** Medicines used to control infections and cure diseases caused by bacteria.
- **Anesthetics:** Substances that cause insensitivity to pain, used during medical operations.
- **Antiseptics:** Substances that prevent the growth of disease-causing microorganisms on living tissue.
- **Chemistry:** The scientific study of the properties, composition, and structure of substances, the transformations they undergo, and the energy changes involved.
- **Chemical products:** Goods manufactured, processed, sold, or distributed that are or contain chemical substances.
- **Chemical industry:** A sector involved in manufacturing and processing chemicals and chemical products.
- **Composition:** The nature of something's ingredients or how a mixture is made up.
- **Disinfectants:** Chemicals used to kill microbes on non-living surfaces.
- **Energy:** The power derived from physical or chemical resources, especially to provide light and heat.
- **Industry:** Economic activity concerned with processing raw materials and manufacturing goods.
- **Insecticides:** Substances used to kill insects.
- **Matter:** Any physical substance that occupies space and has mass.
- **Metallurgy:** (Not explicitly defined in the provided text but listed as a key term; often related to processing metals, which involves chemistry).
- **Property:** An attribute, quality, or characteristic of a substance.
- **Structure:** The arrangement and relationships between the parts or elements of something complex.
- **Substance:** A particular kind of matter with uniform properties.
- **Tranquilizers:** Drugs used to reduce tension and anxiety, bringing calm.

Unit Summary

Chemistry is a fundamental science that explores the properties, composition, and structure of substances, the transformations they undergo, and the associated energy changes. It is divided into five main disciplines: Organic, Inorganic, Physical, Analytical, and Biochemistry. Chemistry's scope is vast, affecting all aspects of life by explaining the natural world, fostering career opportunities, and contributing to agriculture, medicine, food production, and building construction.

Often called the "central science," chemistry forms crucial interconnections with other natural sciences like Biology (Biochemistry), Physics (Physical Chemistry/Chemical Physics), and Geology (Geochemistry), as well as Medicine (Medicinal Chemistry). It is essential for meeting basic needs like food, clothing, shelter, health, energy, and clean air, water, and soil. However, chemistry also carries the responsibility of addressing negative environmental impacts, such as climate change, caused by the production of toxic substances.

Chemistry plays a vital role in industrial advancement. Chemical industries produce a wide array of products, from basic chemicals to finished goods like plastics, drugs, soaps, paints, and agricultural chemicals. Ethiopia has a growing chemical industry sector, with numerous medium and large-scale enterprises manufacturing various chemicals and chemical products, including aluminum sulphate, caustic soda, cement, pharmaceuticals, and consumer goods like soap and cosmetics.

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Answer Key

Question #	Correct letter	Subtopic
1	B	1.1.1 Definition of Chemistry
2	C	1.1.1 Definition of Chemistry
3	A	1.1.1 Definition of Chemistry
4	D	1.1.1 Definition of Chemistry
5	B	1.1.1 Definition of Chemistry
6	A	1.1.2 Scope of Chemistry

7	D	1.1.2 Scope of Chemistry
8	B	1.1.2 Scope of Chemistry
9	C	1.1.2 Scope of Chemistry
10	A	1.1.2 Scope of Chemistry
11	D	1.1.2 Scope of Chemistry
12	B	1.2 Relationship with Other Natural Sciences
13	D	1.2 Relationship with Other Natural Sciences
14	A	1.2 Relationship with Other Natural Sciences
15	C	1.2 Relationship with Other Natural Sciences
16	D	1.2 Relationship with Other Natural Sciences
17	B	1.3 A. Agriculture
18	C	1.3 A. Agriculture
19	A	1.3 A. Agriculture
20	D	1.3 A. Agriculture
21	B	1.3 B. Food Production
22	C	1.3 B. Food Production
23	A	1.3 B. Food Production
24	C	1.3 C. Medicine
25	D	1.3 C. Medicine
26	B	1.3 C. Medicine
27	A	1.3 C. Medicine
28	C	1.3 C. Medicine
29	A	1.3 D. Building Construction Materials
30	B	1.3 D. Building Construction Materials
31	D	1.3 D. Building Construction Materials
32	C	1.4.1–1.4.3 Chemical Industries, Categorization and Products
33	B	1.4.1–1.4.3 Chemical Industries, Categorization and Products
34	D	1.4.1–1.4.3 Chemical Industries, Categorization and Products
35	C	1.4.1–1.4.3 Chemical Industries, Categorization and Products
36	B	1.4.4 Chemical Industries and Products in Ethiopia
37	C	1.4.4 Chemical Industries and Products in Ethiopia
38	A	1.4.4 Chemical Industries and Products in Ethiopia
39	D	1.4.4 Chemical Industries and Products in Ethiopia
40	D	1.4.4 Chemical Industries and Products in Ethiopia

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Placement Table

Subtopic	Question numbers	Count (required)
1.1.1 Definition of Chemistry	1–5	5 (5)
1.1.2 Scope of Chemistry	6–11	6 (6)
1.2 Relationship with Other Natural Sciences	12–16	5 (5)
1.3 A. Agriculture	17–20	4 (4)
1.3 B. Food Production	21–23	3 (3)
1.3 C. Medicine	24–28	5 (5)
1.3 D. Building Construction Materials	29–31	3 (3)
1.4.1–1.4.3 Chemical Industries, Categorization and Chemical Products	32–35	4 (4)
1.4.4 Chemical Industries and Products in Ethiopia	36–40	5 (5)
Total	**1–40**	**40 (40)**

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